

Experiment 5.1

Impulse Response of the System

Program:

```
% Impulse Response of the System
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Sample index
```

```
n = 0:10;
```

```
% Unit impulse input
```

```
x = [1 zeros(1,10)];
```

```
% System:  $y(n)=x(n)+0.5x(n-1)$ 
```

```
b = [1 0.5];
```

```
a = [1];
```

```
% Impulse response
```

```
h = filter(b,a,x);
```

```
% Display result
```

```
disp('Impulse Response h(n):');
```

```
disp(h);
```

```
% Plot
```

```
figure;
```

```
stem(n,h,'filled');
```

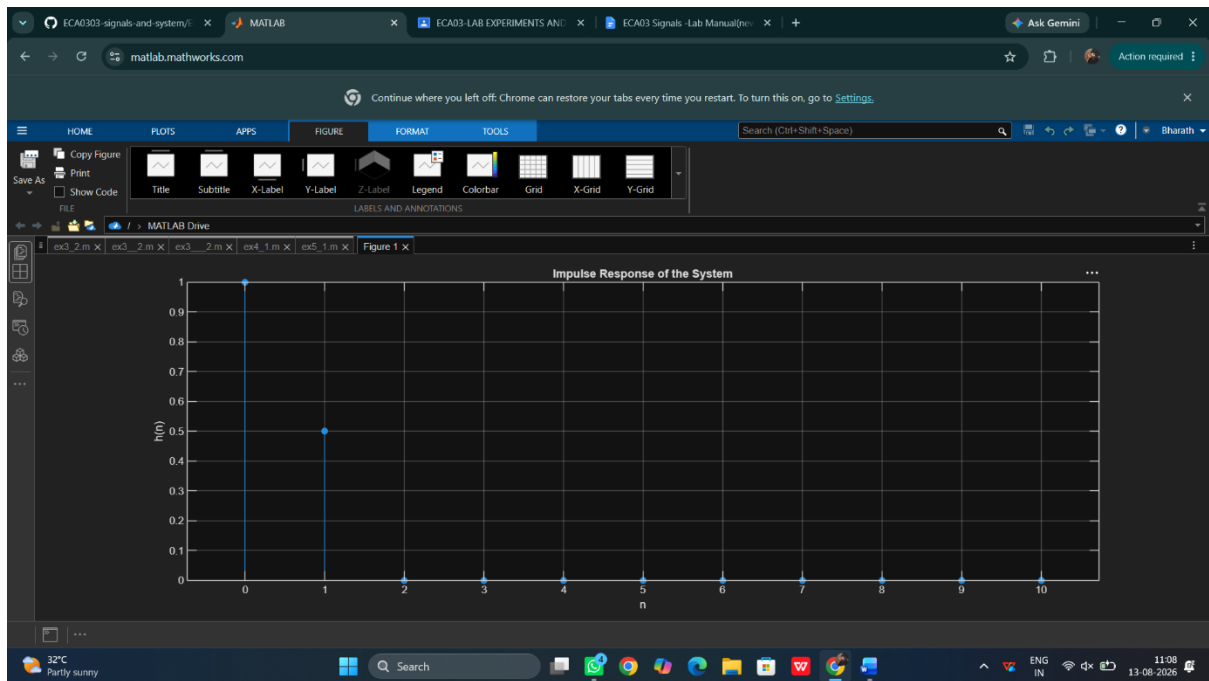
```
grid on;
```

```
title('Impulse Response of the System');
```

```
xlabel('n');
```

```
ylabel('h(n)');
```

Output Waveform:



Result:

The impulse response of the discrete-time system $y(n)=x(n)+0.5x(n-1)$ was obtained by applying a unit impulse signal as input. The resulting impulse response was $h(n)=\{1,0.5,0,0,\dots\}$, which characterizes the behavior of the system.